الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
2011-1432

College of Computing and Informatics

Introduction to Database
DS350

Project

Deadline: Monday 09/08/2026 @ 23:59

[Total Mark is 20]

Student Details: **CRN:50300**

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Instructions:

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, convert text into image or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instructions

- You can work on this project as a group (minimum 3 and maximum 4 students). Each group member must submit the project individually with all group member names mentioned on the cover page.
- This project **worth 20 marks** and will be distributed as in the following:
  - Database Design. **(6 marks)**
  - Database Implementation. **(6 marks)**
  - SQL Operations. **(6 marks)**
  - Project Reflection. **(2 marks)**
- Each leader in the group must submit one report about their chosen Project via the Blackboard **(Email submission will not be accepted and will be awarded ZERO marks)** containing the following:
  A. Task 1 : database design
  B. Task 2: database implementation
  C. All requested queries/results
  D. Screenshots from MySQL (or any other software you use) of **all the tables** after population and **query results**.
  E. Reflection (300–500 words) describing the team's experience, challenges, design decisions, and AI usage (if applicable).

- Submit one **SQL file** containing the complete database implementation, including database creation, tables, data insertion, required queries, VIEW, and TRIGGER.
- Submit a **GitHub** (or GitLab) repository containing the complete project with regular commits and meaningful commit messages.
- Submit a **progress report** describing the completed work, key decisions, challenges, future plan, and contribution of each group member.
- Include **links** to the live report document, GitHub/GitLab repository, and Google Colab notebook (if applicable) in the **Project Artifacts** section of the final report.
- You are advised to make your work clear and well presented; marks may be reduced for poor presentation. This includes filling in your information on the cover page.
- You MUST show all your work, and text **must not** be converted into an image unless specified otherwise by the question.

  - **Late submission will result in ZERO marks being awarded.**
  - The work should be your own. Copying **from students or other resources, including AI software, will result in ZERO marks.**

Learning Outcome(s):

CLO1, CLO2, CLO3, CLO4, CLO5

Project Description

Design and Implementation of a Smart Clinic Database System

A private clinic currently manages its patient information manually, making it difficult to organize appointments, treatments, and payments. The clinic has decided to develop a database system to improve data organization and retrieval.

Your team is required to design, implement, and test a complete relational database using the concepts learned throughout this course. The project should demonstrate your understanding of database design, ER/EER modeling, relational mapping, SQL implementation, and advanced SQL queries. The project must be completed using MySQL.

Task 1 – Database Design (6 Marks)

Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.

A complete ER/EER diagram.

Primary and foreign keys.

Appropriate relationships with cardinality.

At least one specialization/generalization (EER feature).

Any assumptions made during the design process.

Task 2 – Database Implementation (6 Marks)

Complete the following tasks using MySQL. Include screenshots of both your SQL code and the corresponding output (results) for each task:

Create the database and all required tables.

Use appropriate data types and constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, etc.).

Insert at least 5 records into each main table.

Task 3 – SQL Operations (6 Marks)

Complete the following tasks using MySQL. For each task, include screenshots of both your SQL code and the corresponding output (results), along with a brief description (one to two sentences) explaining the purpose of the query and the information it is intended to retrieve.

Write SELECT statements.

Write JOIN queries.

Write nested queries.

Use aggregate functions with GROUP BY.

Write UPDATE and DELETE statements.

Create one VIEW.

Create one TRIGGER.

Task 4 – Reflection (2 Marks)

Write a **300–500-word reflection** on your experience while completing this project by answering the following questions. Your responses should be clear, concise, and supported with specific examples from your own work.

Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.

Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.

If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.

What would you do differently if you were to redo this project with more time or resources?

Deliverables

Project Report (PDF) – System description, ER/EER diagram, relational schema, SQL code, screenshots, and reflection.

SQL Script (.sql) – Complete SQL implementation of the project.

GitHub Repository – Repository link with the complete project and commit history.

Mid-Project Progress Report – Summary of progress, challenges, decisions, and team contributions.

Project Artifacts

Live report document (with edit history): [link]

GitHub/GitLab repository (with commit history): [link]

Google Colab notebook (if applicable): [link]

Smart Clinic Database System

Introduction

A private clinic currently manages patient information manually, which makes it difficult to organize appointments, treatments, medicines, and payment records efficiently. As the amount of information grows, manual record management can lead to data duplication, errors, and delays when retrieving important information. To improve the management of clinic data, a database system is needed to store and organize information in a structured and reliable way.

This project focuses on the design and implementation of a Smart Clinic Database System. The system is intended to support the clinic's daily operations by managing patient records, doctor information, appointments, treatments, medicines, and payments. Using a relational database allows information to be stored consistently while making it easier to access and manage when needed.

Throughout this project, database concepts learned in the course are applied, including ER and EER modeling, relational schema design, SQL implementation, and advanced SQL operations. The database will be implemented using MySQL and will include tables, relationships, constraints, queries, a view, and a trigger. The completed system aims to provide an organized and efficient solution for managing clinic information while demonstrating the practical application of database design principles.

Task 1: Database Design

Private clinics handle a large amount of information related to patients, doctors, appointments, treatments, medicines, and payments. Managing this information manually can make it difficult to maintain accurate records, organize data efficiently, and retrieve information when needed. To address these challenges, a database design is required to provide a structured and organized approach to data management.

For this task, we designed a Smart Clinic Database System using Entity Relationship (ER) and Enhanced Entity Relationship (EER) modeling techniques. The design identifies the main entities involved in the clinic's operations, defines their attributes, and establishes the relationships between them. Primary keys, foreign keys, and cardinality constraints were incorporated to ensure data integrity and accurately represent real-world clinic processes. In addition, specialization and generalization concepts were applied in the EER model to provide a more comprehensive representation of the system. The completed design serves as the foundation for the database implementation phase.

– Database Entities

The ER model of the Smart Clinic Database System consists of the following entities:

- Patients
- Doctors
- Appointments
- Treatments
- Medicines
- Payments

Each entity contains attributes relevant to its role within the clinic and is uniquely identified by a primary key.

To enhance the database design, we introduced a superclass named **PERSON** in the EER model to represent the common attributes shared by Patients and Doctors. The specialization uses an overlapping constraint, allowing an individual to be both a Patient and a Doctor simultaneously, and a partial completeness constraint, allowing a Person to belong to neither subclass. This approach reduces redundancy and provides a more realistic representation of the clinic environment.

– ER Diagram

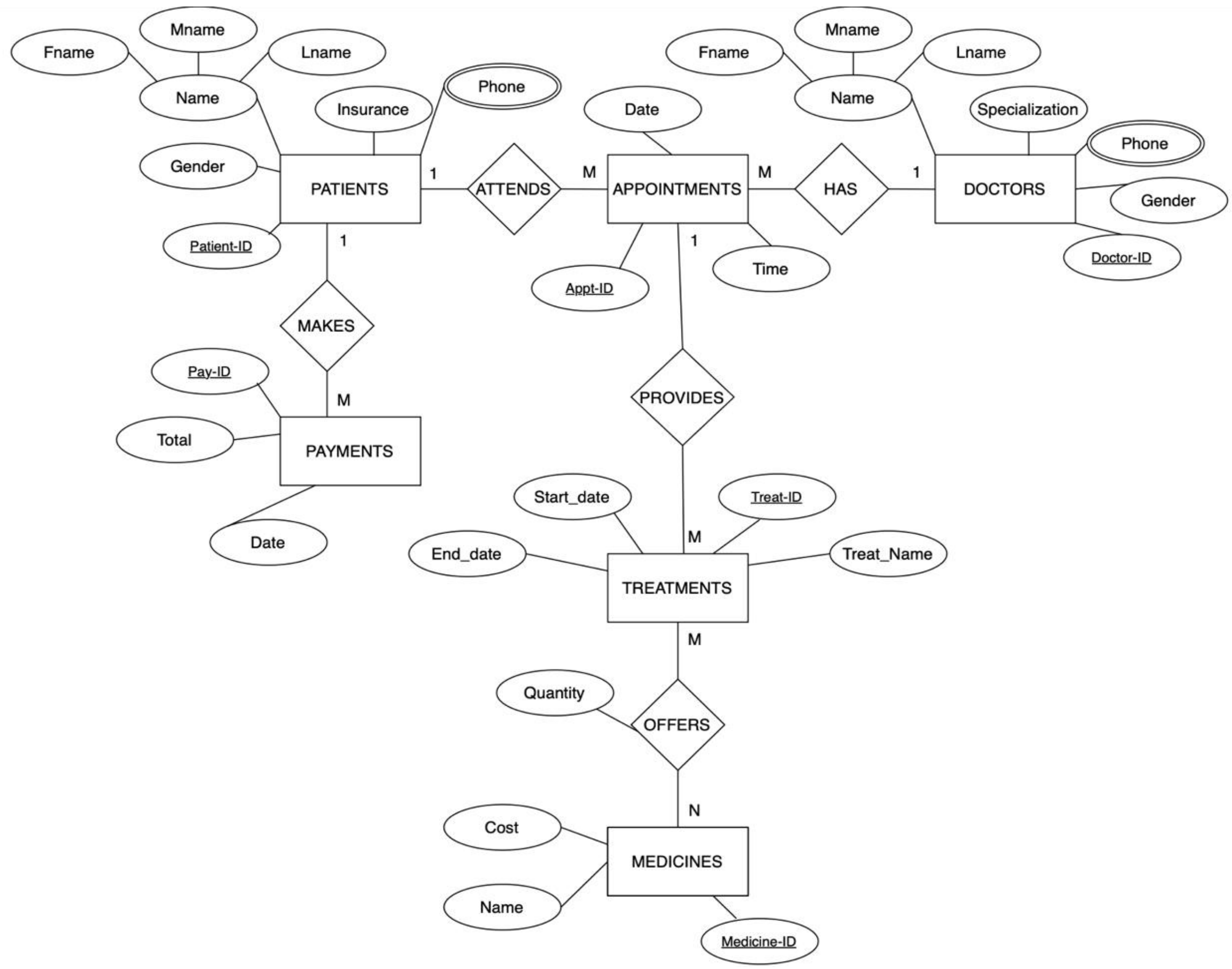

Figure 1: ER Diagram of the Smart Clinic Database System.

The ER diagram illustrates the main entities, attributes, primary keys, relationships, and cardinality constraints of the Smart Clinic Database System. It provides a conceptual representation of the data required to manage patients, doctors, appointments, treatments, medicines, and payments within the clinic.

– EER Diagram

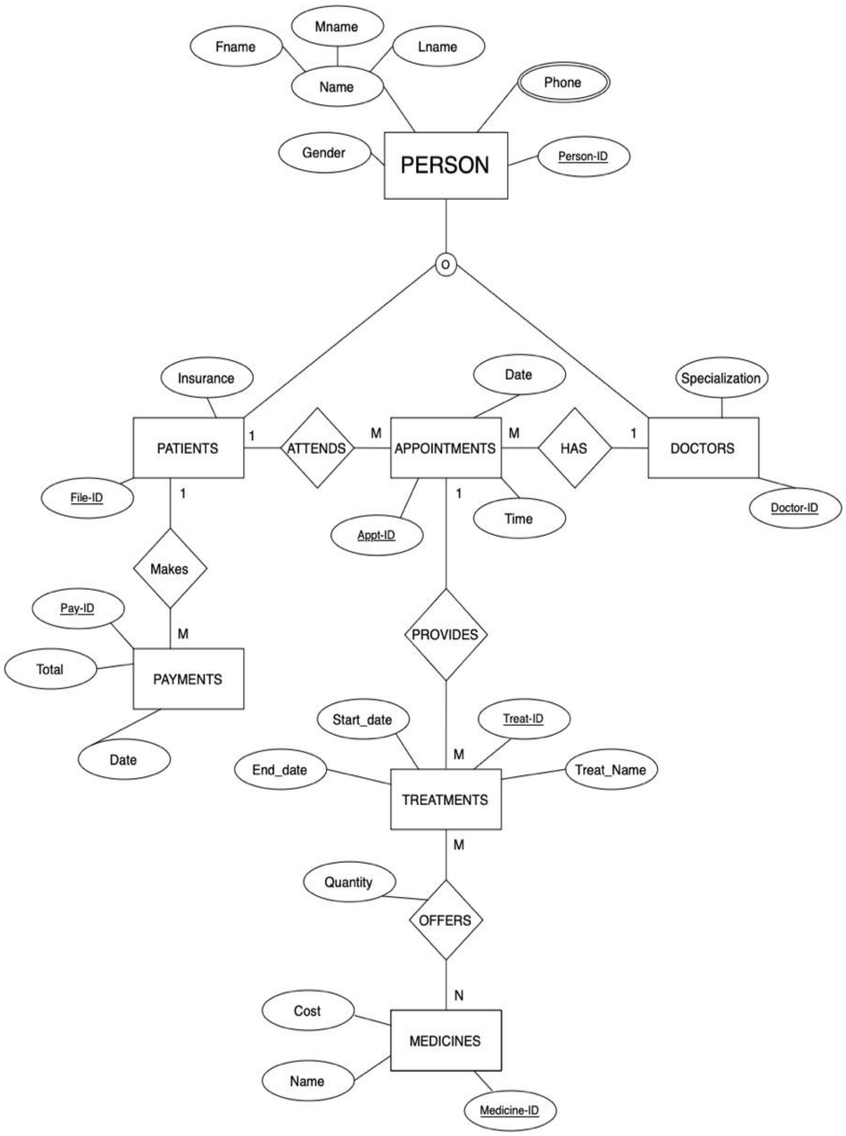

Figure 2: EER Diagram of the Smart Clinic Database System.

The EER diagram extends the ER model by applying the specialization/generalization concept. A Person superclass was introduced to represent common attributes shared by Patients and Doctors, while overlapping and partial specialization constraints were used to model the relationship between the superclass and its subclasses.

– Relationships and Cardinalities

We established several relationships between the entities to accurately represent the clinic's operations and ensure proper data organization.

- **Patient and Appointment**

  A patient can have multiple appointments, while each appointment is associated with one patient. Therefore, a one-to-many (1:M) relationship exists between Patients and Appointments.

- **Doctor and Appointment**

  A doctor can manage multiple appointments, while each appointment is assigned to one doctor. Therefore, a one-to-many (1:M) relationship exists between Doctors and Appointments.

- **Patient and Payment**

  A patient can make multiple payments, while each payment belongs to one patient. Therefore, a one-to-many (1:M) relationship exists between Patients and Payments.

- **Appointment and Treatment**

  An appointment may involve multiple treatments through the **Provide** relationship. Each treatment record is associated with a specific appointment.

- **Treatment and Medicine**

  A treatment may include multiple medicines, and a medicine may be used in multiple treatments. Therefore, a many-to-many (M:N) relationship exists between Treatments and Medicines through the **Offers** relationship.

  The quantity of medicine prescribed is stored as an attribute of the Offers relationship because the quantity may vary depending on the treatment in which the medicine is used.

– Design Assumptions

Several assumptions were made during the database design process.

We assumed that the Patient entity includes an attribute called **Insurance**, while the Doctor entity includes a **Specialization** attribute. In addition, **Phone** was modeled as a multivalued attribute for both Patient and Doctor entities. The primary key of each entity is identified by underlining the corresponding key attribute.

We also assumed that each Appointment may involve multiple Treatments, represented by the Provide relationship. Furthermore, each Treatment may include multiple Medicines, and the quantity of each medicine may vary depending on the prescribed treatment. Therefore, the quantity attribute was associated with the Offers relationship between Treatments and Medicines.

For the EER diagram, we applied the generalization/specialization concept. We assumed that every Person in the superclass may be a Patient, a Doctor, or neither of them. Consequently, a partial completeness constraint was used, represented by a single line connecting the superclass to the specialization.

Additionally, we assumed that an individual may simultaneously be both a Patient and a Doctor. Therefore, an overlapping constraint, represented by the symbol "o", was used instead of a disjoint constraint.

– Primary Keys

The following primary keys were used to uniquely identify records within the database:

Entity	*Primary Key*
PERSON	PersonID
PATIENTS	FileID
DOCTORS	DoctorID
APPOINTMENTS	ApptID
TREATMENTS	TreatID
MEDICINES	MedicineID
PAYMENTS	PayID

These primary keys ensure that each record can be uniquely identified and support the establishment of relationships among entities within the database.

– Summary

In this task, we designed a database for the Smart Clinic Database System using ER and EER modeling techniques. The design includes the required entities, attributes, relationships, primary keys, cardinality constraints, and specialization/generalization concepts needed to support clinic operations. The ER model provides the foundation of the database structure, while the EER model enhances the design through the use of a Person superclass and overlapping specialization for Patients and Doctors. This design will serve as the basis for the implementation and testing phases of the project using MySQL.

Task 2: Database Implementation

Task 3: SQL Operations

Task 4: Reflection

To be completed during the final project phase.

Mid-Project Progress Report

Smart Clinic Database System

- ## Group Members
  - Sara Khalid Alnwaihel
  - Lena Turki Alquraini
  - Jana Mousa Haqawi
  - Mayar Abdulrahman Alshehri

- ## Work Completed During This Period

During this period, our group focused on the database design phase of the Smart Clinic Database System. We analyzed the project requirements and identified the main entities required for the database, including Patients, Doctors, Appointments, Treatments, Medicines, and Payments.

We completed the ER diagram by defining the entities, attributes, primary keys, relationships, and cardinality constraints. We also developed an EER diagram by introducing a Person superclass and applying specialization/generalization concepts. The EER model uses overlapping and partial specialization constraints to represent the relationship between Patients and Doctors.

Additionally, we documented the design assumptions and completed Task 1 of the project report.

- ## Key Decision Made

One important decision made during this phase was introducing a Person superclass in the EER model. Since Patients and Doctors share common attributes such as name, gender, and phone number, using generalization helped reduce redundancy and improve the database structure.

We also decided to use an overlapping specialization because a person may be both a patient and a doctor at the same time.

• Problem Encountered

One challenge we encountered was determining how to model the relationship between Treatments and Medicines. Initially, it was unclear where the quantity of prescribed medicine should be stored.

After reviewing the database design requirements, we decided to store Quantity as an attribute of the Offers relationship because the quantity can vary depending on the treatment. This approach provides a more accurate representation of the clinic's operations.

• Plan for the Next Period

During the next phase of the project, we plan to implement the database using MySQL. The upcoming tasks include:

- Creating the database and all required tables.
- Defining primary keys, foreign keys, and constraints.
- Inserting sample records into each table.
- Writing and testing SQL queries.
- Creating the required VIEW and TRIGGER.
- Capturing screenshots of implementation and query results.
- Completing the reflection section and final project report.

• Group Member Contributions

Group Member	*Contribution*
Sara Alnwaihel	Responsible for Task 1: Database Design, including the ER and EER diagrams and design assumptions.
Lena Alquraini	Responsible for Task 2: Database Implementation using MySQL.
Jana Haqawi	Responsible for Task 3: SQL Operations, including queries, VIEW creation, and TRIGGER implementation.
Mayar Alshehri	Responsible for Task 4: Reflection, report organization, and project coordination.

Project Artifacts

Live Report Document: [Google Doc Smart Clinic Database System Report**]**

GitHub/GitLab Repository: [DS350 Smart Clinic Database System GitHub Repository**]**